

Aman Kumar

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Education

Bachelor of Technology (B.Tech) – Computer Science and Engineering - Indian Institute of Technology (IIT) Patna
2024 - 2028 CGPA: **8.87 / 10**

Experience

Incoming Software Engineering Intern – Google **Summer 2027**

- Selected as a Software Engineering Intern at Google for the Summer 2027 internship program.
- Will contribute to large-scale software engineering projects, collaborating with experienced engineers to design, develop, test, and deploy production-quality software.
- Will work on real-world engineering problems involving scalable systems, software development, debugging, and performance optimization.

Technical Skills

- **Languages:** C++, Python, JavaScript, SQL
- **Frameworks Technologies:** React, Node.js, Express.js, Socket.IO
- **Databases:** PostgreSQL, MongoDB, Supabase
- **Core CS:** Data Structures Algorithms, Object-Oriented Programming, Database Management Systems, Computer Architecture, Digital Logic Computer Organization, Artificial Intelligence Fundamentals
- **Tools:** Git, GitHub, Docker, Vercel, VS Code, Jupyter Notebook

Projects

CP Tracker AI – Full-Stack Competitive Programming Analytics Platform

 GitHub  Live Website

- Built and deployed a full-stack competitive programming analytics platform used by **100+ students** for tracking Codeforces and LeetCode performance.
- Developed rating visualization, contest analytics, and topic-wise weakness detection dashboards using real-time competitive programming data.
- Implemented secure authentication using Google OAuth, Supabase Auth, PostgreSQL, and Row-Level Security (RLS).
- Integrated Gemini AI to generate personalized coaching insights, weak-topic detection, and practice recommendations.

SIMPLEX Assembler Emulator

 GitHub

- Developed a two-pass assembler supporting symbol resolution, instruction decoding, PC-relative addressing, and object code generation.
- Implemented efficient symbol table management and opcode processing for translating SIMPLEX assembly programs into machine code.
- Built a machine emulator capable of executing generated object code with execution tracing and debugging support.
- Generated listing and object files while simulating stack operations and instruction execution.

BiteGo – Food Delivery Platform (Group Project)

 GitHub

- Designed and managed a scalable PostgreSQL database schema using Prisma ORM for users, vendors, orders, and delivery workflows.
- Built a real-time messaging service using Node.js, Socket.IO, MongoDB, and Mongoose supporting persistent conversations and media sharing.
- Implemented token-based authentication and integrated chat services with the main platform database for secure user management.
- Developed backend services for reliable communication and real-time order interactions.

Achievements

- **ICPC Prelims 2025:** AIR **384**, Ranked **15th at IIT Patna**
- **CodeChef:** 3-Star Coder (Max Rating: **1762**)
- **Codeforces:** Specialist (Max Rating: **1488**), solved **950+** algorithmic problems
- **LeetCode:** Knight (Rating: **2065**), solved **610+** problems out of which **140+ Hard**
- **JEE Advanced 2024:** CRL **4342**
- **JEE Main 2024:** **99.47** Percentile

Position of Responsibility

Sub-Coordinator – OptimatX Club (Mathematics)

Organizer – Integral Cup 2026 (Season 1)

- Organized and coordinated Integral Cup 2026 Season 1 at IIT Patna.
- Managed problem setting, contest logistics, participant coordination, and smooth execution of the event.
- Collaborated with club members and volunteers to ensure fair evaluation and an engaging contest experience.

Languages Interests

- **Languages:** English, Hindi
- **Interests:** Competitive Programming, Algorithms, Mathematics, Software Engineering